LearningMore - File Management

Owner:
Reviewer:
Contributors:
Date Generated: Mon Apr 20 2026

Executive Summary

High level system description

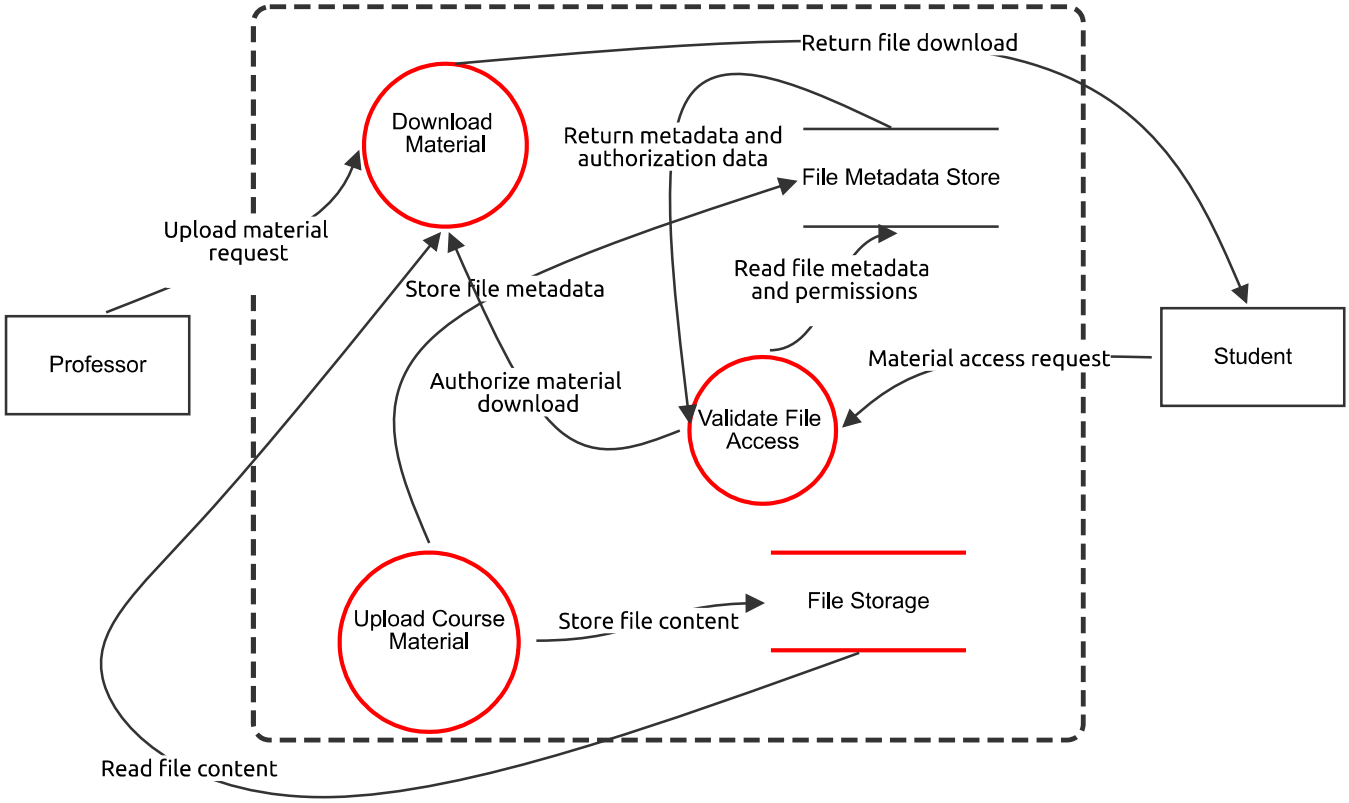
Threat model for the File Management component of the LearningMore platform, including upload of course materials, authorization checks, and download of protected files.

Summary

Total Threats	12
Total Mitigated	0
Total Open	12
Open / Critical Severity	4
Open / High Severity	6
Open / Medium Severity	2
Open / Low Severity	0

File Management DFD Level 2

DFD for File Management including upload, authorization validation, metadata handling, and file download.



File Management DFD Level 2

Professor (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Student (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Upload CourseMaterial (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Malicious file upload	Tampering	High	Open		User uploads malicious file disguised as valid content	File type validation + antivirus scanning + size limits
3	Unauthorized upload	Elevation of privilege	High	Open		Student attempts to upload materials without permission	RBAC + role validation (only professors allowed)
4	Large file upload attack	Denial of service	Medium	Open		User uploads excessively large files to exhaust storage/resources	File size limits + rate limiting

Validate File Access (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
14	Forged authentication token	Spoofing	Critical	Open		Attacker uses forged JWT to bypass access control	JWT validation (signature, expiry, issuer)
15	Authorization bypass	Elevation of privilege	Critical	Open		User accesses files without enrollment or permission	Strict server-side authorization checks
16	Unauthorized metadata access	Information disclosure	High	Open		Attacker retrieves file metadata without permission	Access control on metadata queries

Download Material (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
8	Unauthorized file download	Information disclosure	Critical	Open		User downloads restricted files without authorization	Access validation before file retrieval
9	Download flooding	Denial of service	Medium	Open		Multiple download requests overload system	Rate limiting + throttling
10	Metadata manipulation	Tampering	High	Open		Attacker alters file permissions or ownership	Integrity checks + restricted DB access
11	Metadata leakage	Information disclosure	High	Open		Sensitive metadata exposed (paths, permissions)	Access control + data minimization

File Metadata Store (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

File Storage (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
12	File modification	Tampering	High	Open		Stored files are altered or replaced	Hash validation + secure storage
13	Direct file access	Information disclosure	Critical	Open		Files accessed directly bypassing application	Store outside public directories + signed URLs

Upload material request (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Store file content (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Store file metadata (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Material access request (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Read file metadata and permissions (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Return metadata andauthorization data (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Authorize material download (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Read file content (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Return file download (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations